AUTONOMOUS CONTROL TOWER • MULTI-AGENT SYSTEM (MAS) ENGINEERING BLUEPRINT

OMNI-LOGISTICS MASTER AGENT

Unifying Terminal Workflows, Customs Pre-Clearance, Proactive Disruption Routing, and Real-Time Margin Optimization into a Single Zero-Cost AI Orchestrator

STRATEGIC VISION: THE "HOLY GRAIL" OF GLOBAL TRADE

SYS-OVERVIEW // DUBAI HUB

Anchored by world-scale port infrastructure such as Jebel Ali and expansive free zones, Dubai operates as a premier global logistics hub. Converting high-velocity operational telemetry into governed actions requires moving beyond fragmented human workflows. In enterprise artificial intelligence, combining terminal digitization, real-time shipment tracking, predictive disruption modeling, and inland coordination into one unified system is known as an **Autonomous Control Tower** or **Orchestrator Agent**. Rather than human operators manually stitching together data across ships, ports, and trucks, the single Omni-Agent sits atop enterprise software (ERP, TMS, WMS) as a central brain. Under the hood, it executes as a **Multi-Agent System (MAS)**—a Master Agent delegating specialized tasks to a **Terminal Agent**, **Customs Agent**, and **Routing Agent**—transforming a fragile chain of manual handoffs into a self-healing, autonomous supply chain network.

Zero-Cost Open-Source Stack ($0/Mo MVP): Frontend: Next.js 14 (App Router, Tailwind, Vercel Free) • **Backend:** Python 3.11 + FastAPI (Render Free) • **Database:** Supabase (Free PostgreSQL) • **Orchestration:** CrewAI / LangGraph • **Reasoning Engine:** Gemini 1.5 Flash (Google AI Studio Free Tier)

SOLVING THE 4 HIGH-COST SUPPLY CHAIN BOTTLENECKS

MAS PROBLEM MATRIX

PROBLEM 01 // PORT-TO-INLAND FRICTION

The "Blind Handoff"

Current Issue: When cargo vessels arrive at massive ports like Jebel Ali, terminal container unloading is disconnected from local trucking fleets. Delayed ships leave trucks waiting empty; early ships leave containers incurring yard storage fees.

Omni-Agent Solution: Synchronizes ship AIS tracking directly with inland trucking schedules. If a vessel is 12 hours late, the agent autonomously cancels gate bookings, reschedules for exact crane clearance, and reassigns drivers to profitable local routes.

PROBLEM 02 // FREE ZONE COMPLIANCE

The Paperwork Chokehold

Current Issue: Moving goods through Dubai Free Zones into the mainland or re-exporting globally demands heavy documentation (Bills of Lading, commercial invoices, certificates of origin). Manual entry causes typos and customs holds.

Omni-Agent Solution: Extracts unstructured data from emails and PDFs via vision and NLP, cross-references UAE customs rules in real time, and submits declarations to Dubai Customs portals before docking for 100% pre-clearance.

PROBLEM 03 // GEOPOLITICAL & WEATHER SHOCKS

Reactive Disruption Management

Current Issue: Supply chains react too slowly when geopolitical events block shipping lanes or severe storms ground flights, leaving teams scrambling for days to locate inventory and book alternative capacity.

Omni-Agent Solution: Continuously ingests global weather, news, and port congestion metrics. Upon detecting a disruption, it runs a Cost vs. Speed simulation, splits cargo, and books air freight for high-value items while rerouting ocean vessels.

PROBLEM 04 // OPERATIONAL VS. FINANCIAL SILOS

Margin Erosion from Siloed Data

Current Issue: Logistics operators lose millions annually because dispatchers cannot see the real-time financial impact of operational choices across fuel surcharges, demurrage, and strict customer SLA delay penalties.

Omni-Agent Solution: Connects live operational telemetry (truck location, berth wait) with financial parameters (fuel cost, SLA penalty curves). Every autonomous reroute or crane-lift upgrade is mathematically optimized to protect profit margins.

1. CORE PLANNING DOCS: PRD & TECHNICAL DESIGN (TRD) DOC-SET // PART 01A

Product Requirements Document (PRD)

- **Core Problem:** Manual supply chain handoffs between marine terminals, customs portals, and inland trucking fleets cause cascading delays, demurrage fees, and severe profit margin erosion.
- **Target User:** Logistics Coordinators and Supply Chain Managers at Third-Party Logistics (3PL) enterprises operating across port and free-zone corridors.
- **Core User Flow:** The backend ingests a simulated port delay event → the Omni-Agent coordinates 3 specialized sub-agents to evaluate customs, berth, and routing alternatives → the Control Tower UI presents an optimized rerouting resolution with cost/time metrics for one-click approval.
- **Out of Scope (MVP):** Paid live integrations with restricted global port APIs, live bank payment rails, and physical warehouse robotics control.

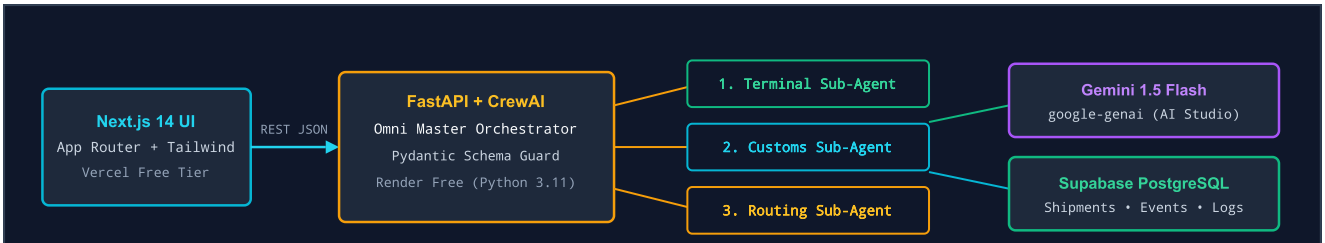


Figure 1. Zero-Cost Autonomous Control Tower Architecture & Sub-Agent Delegation Pipeline

TECHNICAL DESIGN (TRD) & MULTI-AGENT ORCHESTRATION STACK & DATA FLOW

The Next.js 14 frontend polls the FastAPI backend for live shipment telemetry and exception alerts. FastAPI hosts the CrewAI orchestration layer and exposes strictly typed REST endpoints validated by Pydantic models. All LLM reasoning executes server-side via the `google-genai` SDK using **Gemini 1.5 Flash**:

- **Master Orchestrator Agent:** Evaluates incoming disruption events, decomposes recovery objectives, delegates analysis to sub-agents, computes aggregate financial margin impact, and enforces confidence governance thresholds.
- **Terminal Sub-Agent:** Processes simulated AIS vessel ETA shifts, crane availability windows, and gate appointment re-slotting.
- **Customs & Routing Sub-Agents:** Cross-check UAE Free Zone HS code declarations for pre-clearance and simulate split-cargo air/ocean rerouting against SLA penalty curves.

Control Tower UI/UX Design Brief

Theme & Tone: High-contrast Dark Mode "Autonomous Control Tower" engineered for rapid anomaly scanning in 3PL operations centers. **Color Palette:** Deep Slate backgrounds (■ `#0F172A`), Neon Cyan (■ `#06B6D4`) for active corridors, Amber (■ `#F59E0B`) for vessel/gate delays, and Emerald (■ `#10B981`) for autonomous customs & route clearances. **Typography:** *Inter* for operational UI density and tables; *JetBrains Mono* for live agent Chain-of-Thought reasoning logs, container IDs, and financial delta metrics.

APP FLOW & CONTROL TOWER SCREEN STATES

DOC-SET // PART 01B

- **Screen 1 — Global Control Tower Dashboard:** Displays active shipments across port-to-inland corridors on an interactive dark-mode map alongside a real-time telemetry feed, container SLA statuses, and current margin-at-risk counters.
- **Screen 2 — Proactive Disruption Alert Modal:** Triggers automatically when the backend simulator injects an anomaly (e.g., *"Vessel MSC Zoe delayed by 12 hours at Jebel Ali Berth 4"*), highlighting affected high-value containers and SLA penalties.
- **Screen 3 — Autonomous Agent Resolution Drawer:** Displays the Omni-Agent's synthesized recovery plan (*"Cancel gate booking A, pre-clear Customs BoL #994, reserve Air Freight B for high-value split"*), comparing net cost vs. time saved, an expandable **Agent Chain of Thought** log, and an **"Execute Autonomous Fix"** action button.

SUPABASE BACKEND SCHEMA & IMPLEMENTATION PLAN

POSTGRESQL & ROADMAP

Table / Milestone	Fields / Engineering Deliverables	Relationships & System Role
`shipments` (Table)	`id (UUID PK), container_id (TEXT), status (TEXT), expected_arrival (TIMESTAMPTZ), value (NUMERIC)`	Primary entity tracking active containers, cargo value, and current transit state.
`events` (Table)	`id (UUID PK), shipment_id (FK), event_type (TEXT), description (TEXT), timestamp (TIMESTAMPTZ)`	Foreign key to `shipments.id`; stores AIS delays, weather alerts, and customs holds.
`agent_decisions` (Table)	`id (UUID PK), shipment_id (FK), sub_agent (TEXT), proposed_action (JSONB), cost_impact (NUMERIC), executed (BOOL)`	Foreign key to `shipments.id`; audits sub-agent proposals, net financial impact, and execution state.
M0 & M1 (Foundation)	**M0:** Monorepo setup, Supabase provisioning, Next.js 14 + FastAPI scaffolding. **M1:** FastAPI-to-Supabase plumbing & synthetic data seeder.	Establishes local/cloud zero-cost pipeline and populates 5 realistic Dubai corridor shipments.
M2 & M3 (MAS + UI)	**M2:** CrewAI Master, Terminal, Customs & Routing agents wired to Gemini 1.5 Flash. **M3:** Next.js dashboard state integration.	Connects backend multi-agent reasoning to live frontend shipment selection and cost/time visualization.
M4–M8 (Loop & Scale)	**M4:** End-to-end delay simulation & resolution write-back. **M5–M8:** HS code RAG, PDF BoL OCR parser, SLA margin simulator & Vercel/Render deploy.	Completes autonomous closed-loop recovery and scales MVP into production hardening milestones.

2. CODING AGENT INSTRUCTION FILE (`AGENTS.MD`)

DOC-SET // PART 02

```
# Omni-Logistics Agent Project Rules (Save as AGENTS.md or .agent/rules/rules.md)
## Tech Stack & Exact Versions
- Frontend: Next.js 14.2+ (App Router), TailwindCSS 3.4+, TypeScript 5+, Lucide Icons.
- Backend: Python 3.11+, FastAPI 0.110+, CrewAI 0.30+, Pydantic 2.6+, google-genai SDK.
- Database & LLM: Supabase (PostgreSQL 15) | Gemini 1.5 Flash via Google AI Studio.
## Folder Structure & Naming Rules
- `/frontend/src/app` (Next.js routes), `/frontend/src/components` (PascalCase UI
components).
- `/backend/app/main.py` (FastAPI entry), `/backend/app/agents/` (CrewAI definitions),
`/backend/app/services/mock_data.py` (synthetic simulator), `/backend/app/schemas.py`
(Pydantic models).
## Strict Architecture & Data Rules
1. NEVER hardcode API keys or DB URLs. Always read `process.env` (Node) or `os.environ`
(Python).
```

3. ACCEPTANCE CRITERIA PER MILESTONE

DOC-SET // PART 03

Milestone	Verification Checklist ("Definition of Done")	Required Test Command / Proof
M0: Setup	Next.js (`:3000`) and FastAPI (`:8000`) run locally without errors. Supabase credentials verified; dummy test row inserted and queried from `shipments`.	`pytest tests/test_db.py` & `GET /health` returns `200 OK`.
M1: Data	`services/mock_data.py` seeds 5 synthetic Jebel Ali shipments. Next.js dashboard fetches and renders all 5 containers with status badges and arrival timestamps.	`GET /api/shipments` returns validated 5-item JSON array.
M2: Agents	Posting a mock 12-hour vessel delay JSON payload triggers the CrewAI Master Agent, which delegates to Terminal, Customs, and Routing sub-agents and returns a Pydantic-validated resolution plan via Gemini 1.5 Flash.	`POST /api/simulate-delay` returns structured action + confidence score.
M3: Integration	Clicking a delayed shipment on the Next.js dashboard triggers the backend CrewAI pipeline and renders proposed cost impact ($), time saved (hrs), source citations, and collapsible Chain-of-Thought reasoning.	UI Alert Modal & Resolution Drawer render with zero console errors.
M4: Completion	Clicking "**Execute Autonomous Fix**" updates `shipments.status` to `RESOLVED` in Supabase and appends an immutable audit record to `agent_decisions` with `executed = true`.	Supabase row update verified + `npm run lint` & `pytest` pass.
M5–M8: Extended	Exporter enters a product description in English and receives 3 synthetic UAE HS code suggestions with cited tariff sources, confidence scores, and automatic escalation when confidence falls below threshold.	End-to-end human-in-the-loop gate verified in UI & API tests.

4. DATA GOVERNANCE, SAFETY & WHEN THE AGENT MAKES A MISTAKE

DOC-SET // PART 04

Human-in-the-Loop Guardrails & Content Rules

- **Synthetic Source Data & Licensing:** Because live commercial logistics APIs (e.g., Project44, MarineTraffic) are proprietary and paid, all MVP vessel AIS telemetry, port weather, buyer records, and UAE customs rules are generated deterministically via `services/mock_data.py` under an open MIT project license.
- **Mandatory Source Attribution:** Every HS code recommendation, customs pre-clearance draft, and rerouting quote must display its exact underlying data source (e.g., `[Source: Mock-UAE-Tariff-2026-Ch84]` or `[Source: Sim-AIS-JebelAli-Berth04]`).
- **Confidence & Financial Thresholds ("What Happens If the Agent Makes a Mistake?"):** The Master Agent evaluates a mathematical confidence score and net financial exposure on every plan. If **Cost Impact > $10,000** OR **Confidence < 85%**, or if tariff data is ambiguous, the agent is strictly forbidden from auto-executing—it must output *"I'm not sure — Human Review Required"*, lock the autonomous execution button, and route the exception to a senior supervisor with a one-click rollback snapshot.
- **Auditability & Chain of Thought:** Every resolution card in the UI includes a collapsible "**Agent Chain of Thought**" drawer exposing the exact sub-agent prompts, SLA penalty math, and variables used by Gemini 1.5 Flash.

5. ENVIRONMENT SECRETS, `.GITIGNORE` & README SETUP

DOC-SET // PART 05

```
# === FILE 1: .env.example (Commit this template; NEVER commit real keys) ===
NEXT_PUBLIC_API_URL=http://localhost:8000
GEMINI_API_KEY=your_google_ai_studio_api_key_here
```